LIFE EXPECTANCY PREDICTION PROJECT USING WORLD BANK DATA

1. Introduction

1.1. Motivation and Objective

The primary objective of this project is to analyze the socio-economic and environmental determinants of human life expectancy through the development of a comprehensive, composite index-based predictive model. As life expectancy serves as a definitive metric for a nation's overall health and developmental progress, understanding the multi-dimensional factors that drive this metric is crucial for policy formulation and resource allocation. This study aims to bridge the gap between raw World Bank socio-economic data and actionable insights by constructing five distinct composite indices: *Health and Immunity*, *Socio-Technological Mobility*, *Environmental Quality*, *Basic Infrastructure and Resources*, and *Social Risk and Justice*.

1.2. Methodological Evolution

Building upon the feedback from previous project phases, this iteration introduces significant methodological refinements to ensure both robustness and academic integrity. A primary concern in previous findings was the potential for circular logic, wherein outcome-based variables (e.g., infant mortality rates) might artificially inflate the model's predictive performance. To address this, the current study has strictly isolated these variables, ensuring that all indices are composed solely of input-based indicators (e.g., vaccination coverage, resource access, and healthcare expenditure). This refinement allows the model to function not merely as a correlation machine, but as a simulator of systemic impacts.

1.3. Scope of Analysis

This project employs a rigorous analytical pipeline, beginning with data normalization and the inversion of negative-impact variables, followed by Principal Component Analysis (PCA) to derive latent developmental dimensions. The resulting composite indices serve as the foundation for various machine learning models, with XGBoost emerging as the optimal framework. Furthermore, this study integrates an ablation analysis to quantify the systemic contribution of each index, ensuring that the model's predictive power is attributed to coherent developmental systems rather than isolated data points. Finally, the inclusion of a simulation tool enables users to assess the potential impact of strategic policy interventions on national life expectancy, positioning this study as a foundation for evidence-based decision-making.

2. Literature Review

2.1. Composite Indices in Developmental Economics

The measurement of national development has shifted from singular metrics, such as GDP per capita, toward multidimensional frameworks. The Human Development Index (HDI) established the foundation for considering health, education, and standard of living as interconnected pillars (UNDP, 1990). Modern literature emphasizes the necessity of composite indices to capture the latent variables of development. Recent studies advocate for Principal Component Analysis (PCA) as an objective weighting mechanism to eliminate the subjectivity inherent in manual index construction, a methodology adopted in this study to ensure statistical rigor.

2.2. Determinants of Life Expectancy: A Systemic Perspective

Life expectancy is widely recognized as a complex outcome of systemic socioeconomic conditions rather than isolated health interventions. Research by Preston (1975) established the non-linear relationship between national income and life expectancy, while contemporary studies highlight the impact of environmental quality and infrastructure (e.g., access to clean water and sanitation) as critical precursors to demographic health. Furthermore, recent literature underscores the role of "digital divide" and technological access—captured here under *Socio-Technological Mobility*—as new determinants of health outcomes in an interconnected global economy.

2.3. Machine Learning in Public Policy Simulation

Predictive modeling in public policy has moved beyond traditional linear regression toward ensemble-based machine learning. Methods such as Random Forest and XGBoost have gained prominence due to their capacity to capture non-linear interactions between developmental variables, which often elude standard econometric models. The use of "what-if" simulations—as demonstrated in this project—aligns with the growing field of Evidence-Based Policy Making (EBPM), where machine learning models serve as sandbox environments for policymakers to test intervention strategies before implementation.

3. Data & Methodology

3.1. Data Acquisition and Preprocessing

The dataset utilized in this study is sourced from the World Bank's comprehensive development indicators. The preprocessing pipeline was designed to handle high-dimensional, longitudinal data:

- **Missing Value Imputation:** To preserve the longitudinal integrity of the dataset, missing values were addressed using a mix of interpolation for time-series consistency and median-imputation for static cross-sectional gaps. This ensured the dataset remained robust without introducing significant bias.
- **Feature Engineering:** Following the constructive critique regarding circular logic, all target-dependent outcome variables (e.g., infant mortality, child survival) were removed.
- **The "Is_Pandemic" Indicator:** We engineered a binary categorical variable (Is_Pandemic) for the year 2020. This inclusion was essential to isolate the systemic

shocks caused by the COVID-19 pandemic, preventing the model from misinterpreting a temporary, exogenous global disruption as a permanent structural shift in life expectancy.

- **Normalization:** All features were scaled using Min-Max scaling to ensure that variables with differing units (e.g., USD, percentages, index points) contributed equitably to the PCA transformation.

3.2. Construction of Composite Indices (PCA)

To synthesize high-dimensional data into intuitive developmental pillars, we employed Principal Component Analysis (PCA). PCA was chosen over arbitrary weight assignment to provide a mathematically objective measure of importance.

3.3. Predictive Modeling and Ablation Study

The project employs a robust multi-model comparison framework. To ensure the optimal predictive performance, we trained and evaluated seven distinct regression models: Linear Regression, Ridge Regression, Lasso Regression, Random Forest, Gradient Boosting, XGBoost, and Support Vector Regression (SVR).

This comparative approach served two purposes:

- **Model Selection:** By evaluating models ranging from simple linear architectures to complex ensemble methods, we were able to empirically determine that tree-based ensemble learners (specifically XGBoost and Random Forest) best captured the non-linear dynamics of developmental data.

- **Validation (K-Fold Cross-Validation):** To prevent overfitting and ensure the model's generalizability across different developmental contexts, we utilized 5-Fold Cross-Validation. This process provided not only the average R^2 scores but also the standard deviation, allowing us to assess the stability of each model.

Following the model selection, an Ablation Study was conducted on our top-performing model (XGBoost). By systematically withholding each composite index, we measured the resulting decline in performance. This analysis confirmed that the predictive power is rooted in a "systemic developmental profile" rather than the influence of any single isolated variable.

3.4. Visualization and Simulation Tools

To ensure the interpretability of our findings, a multi-layered visualization strategy was employed:

- **Seaborn and Matplotlib:** These libraries were instrumental in mapping the relationships between composite indices and the target variable through regplot-based scatter analysis.

- **Correlation Heatmaps:** Used to identify multicollinearity between the constructed indices, verifying the systemic nature of our variables.

- **Bar Charts (Feature Importance):** These were utilized to communicate the weight of each index, ensuring that our model's logic remains transparent and easy to scrutinize.
- **Simulation Engine:** The application layer includes an interactive simulation tool, transforming our static predictive model into a dynamic "what-if" sandbox for evidence-based policy testing.

4. Results and Evaluation

4.1. Model Benchmarking and Performance

The initial phase of our analysis involved a comparative evaluation of seven regression models. As summarized in the table below, tree-based ensemble methods significantly outperformed linear architectures, confirming the presence of complex, non-linear relationships within the developmental indicators.

Model	Mean R^2 Score	Standard Deviation
XGBoost	**0.9578**	**0.0029**
Random Forest	0.9502	0.0039
Gradient Boosting	0.8711	0.0119
SVR	0.7671	0.0140
Linear Regression	0.6835	0.0157
Ridge Regression	0.6835	0.0156
Lasso Regression	0.3863	0.0100

XGBoost was identified as the optimal model, exhibiting the highest predictive accuracy and the lowest standard deviation, indicating stable performance across different developmental contexts.

4.2. Ablation Analysis: Evaluating Systemic Integrity

To address the inquiry regarding the individual contribution of our composite indices and to rule out circular logic, an ablation study was performed. The results demonstrate that no single index acts as a "crutch" for the model.

As shown above, the exclusion of any single index—particularly *Socio-Technological Mobility* and *Basic Infrastructure*—leads to a substantial drop in predictive performance (R^2 dropping from ~0.95 to ~0.40). This validates that the model does not rely on a single dominant variable; rather, it identifies life expectancy as an emergent property of a systemic developmental profile.

4.3. Feature Importance and PCA Loadings

The PCA-based weighting mechanism objectively identified the drivers within each index. For instance, the *Temel_Altyapi_Kaynak* (Basic Infrastructure) index was most heavily influenced by clean water access and sanitation efficiency, confirming that these foundational requirements remain the primary predictors for life expectancy. The high correlation between *Socio-Technological Mobility* and *Basic Infrastructure* (0.69) further supports the systemic hypothesis, as countries with superior physical infrastructure consistently demonstrate higher technological adoption.

4.4. Policy Simulation Insights

The simulation engine validates the model's practical utility. When simulating a 10% improvement in health and infrastructure indicators for the case of Türkiye (2019), the model projects a positive impact on life expectancy. While the marginal gain appears incremental, it provides a quantitative baseline for policy prioritization, demonstrating the model's capacity to translate developmental targets into tangible demographic outcomes.

5. Deployment and Infrastructure Architecture

5.1. Model Serialization

The final trained XGBoost model was serialized using the joblib library. This process converts the in-memory Python object into a persistent file (yasam_beklentisi_model.pkl), ensuring high performance and portability. This allows the model to be instantly loaded into production environments without the computational overhead of retraining.

5.2. API Integration and Model Serving

To transform the static predictive model into a dynamic service, we implemented a RESTful API using FastAPI. This framework was chosen for its high concurrency, asynchronous support, and automatic interactive API documentation (Swagger UI). The API exposes a /predict endpoint, which accepts JSON-formatted socio-economic indicators and returns the calculated life expectancy prediction. This architecture ensures that the model can be seamlessly integrated into web dashboards or mobile policy-simulation applications.

5.3. Security Considerations

In alignment with enterprise-grade deployment standards, the following security measures are planned for a production-level rollout. It should be noted that while a local tunneling prototype was developed for validation, public-facing deployment was deferred in this assignment to adhere to strict corporate security standards. Production requirements include:

- **Authentication:** Implementing OAuth2 or API key management to prevent unauthorized model access.
- **Infrastructure Security:** Utilizing an API Gateway to handle rate limiting and traffic management, mitigating potential Denial-of-Service (DoS) risks.
- **Environment Isolation:** Deploying the API within a containerized environment (e.g., Docker) to ensure consistency across development, testing, and production stages.

5.4. Monitoring and Logging

Effective deployment requires continuous visibility into model health. The proposed architecture includes:

- **Logging:** Capturing all incoming API request payloads, prediction outputs, and system errors into centralized log files for post-hoc analysis.
- **Drift Detection:** Monitoring input feature distributions to identify "data drift"—a scenario where the statistical properties of real-world data deviate from the training dataset, necessitating a model update.

6. Conclusion

This study has successfully demonstrated that human life expectancy is not an isolated demographic metric, but an emergent property of complex socio-economic and environmental systems. By constructing five custom composite indices—*Health and Immunity, Socio-Technological Mobility, Environmental Quality, Basic Infrastructure and Resources,* and *Social Risk and Justice*—and employing a machine learning-based analytical pipeline, we have moved beyond traditional predictive modeling toward a systemic understanding of development.

The methodological rigor introduced in this phase, particularly the Ablation Study, provided compelling evidence against the concerns of circular logic. Our findings confirm that the model's high predictive accuracy is derived from the integration of diverse developmental inputs rather than the reliance on definitionally linked outcome variables. Furthermore, the objective weighting of indicators through Principal Component Analysis (PCA) ensures that our findings are grounded in mathematical transparency rather than subjective assumptions.

The developed Simulation Engine stands as the project's most significant practical contribution. By enabling "what-if" policy scenarios, this tool bridges the gap between data science and evidence-based decision-making. Future iterations of this work could expand on this foundation by incorporating regional-level data or integrating additional exogenous variables such as geopolitical stability or climate risk indices, further enhancing the model's granularity.

In summary, this research provides a scalable, reproducible, and transparent framework for analyzing the drivers of human longevity. It underscores that while technology and infrastructure are essential, their true impact on demographic health is realized only when integrated into a cohesive, system-wide developmental strategy.

Kübra Tuğçe Kızılboğa

Feyza Nur Aydınoğlu